DATA 308: Lecture 9b

Logistic regression

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Examples of Outcome Variables

Specify variable type (think about the possible values)

- Temperature
- Presence of Alzheimer's
- Rent
- Computer operating system
- Daily number of customers
- Income class

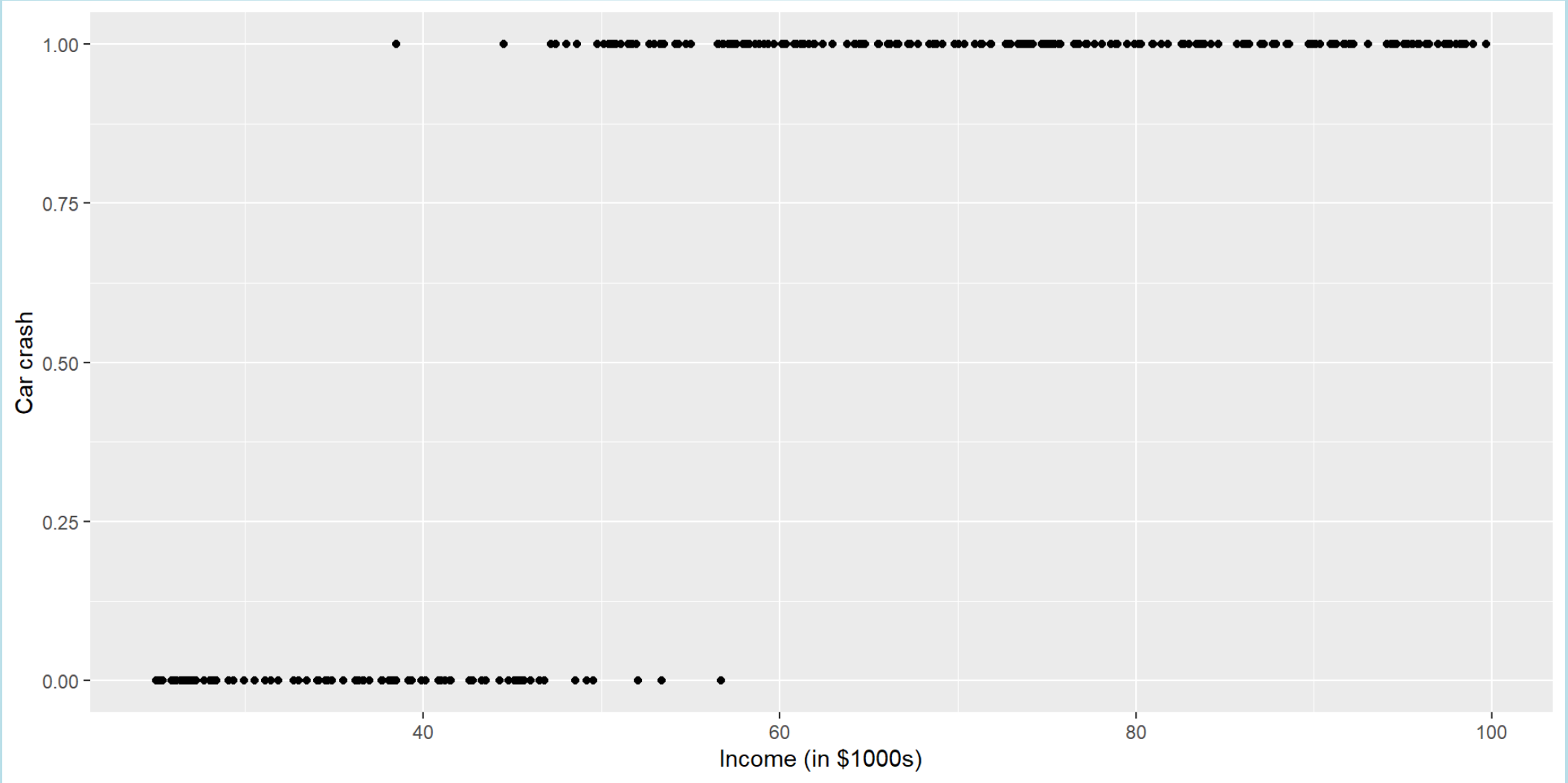
Which could we model with **linear regression**?

Identify the response variable type:

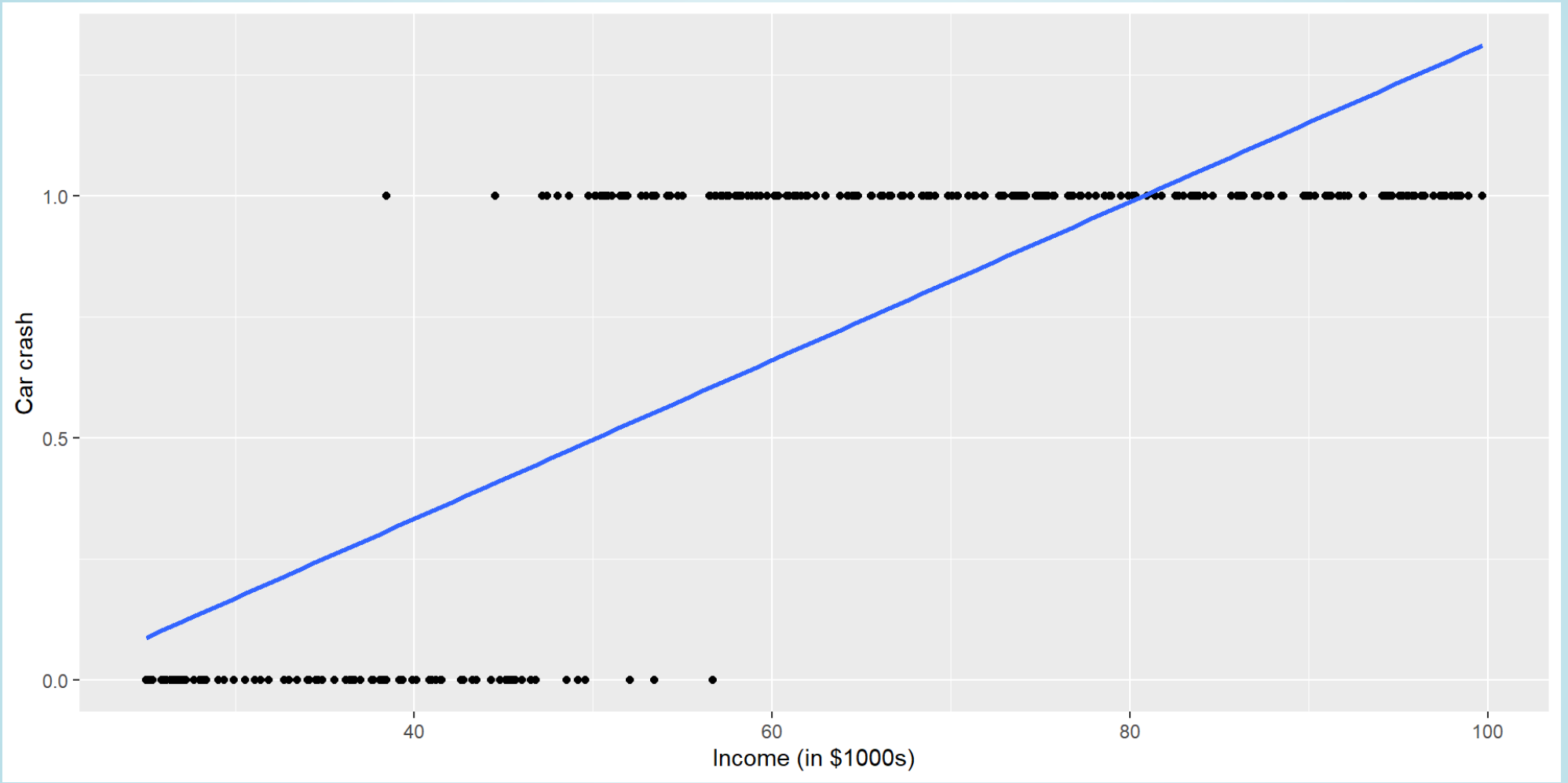
- A car insurance company is trying to predict if an applicant will have a car crash in the next 5 years, based on their income.

Can we model using linear regression anyways?

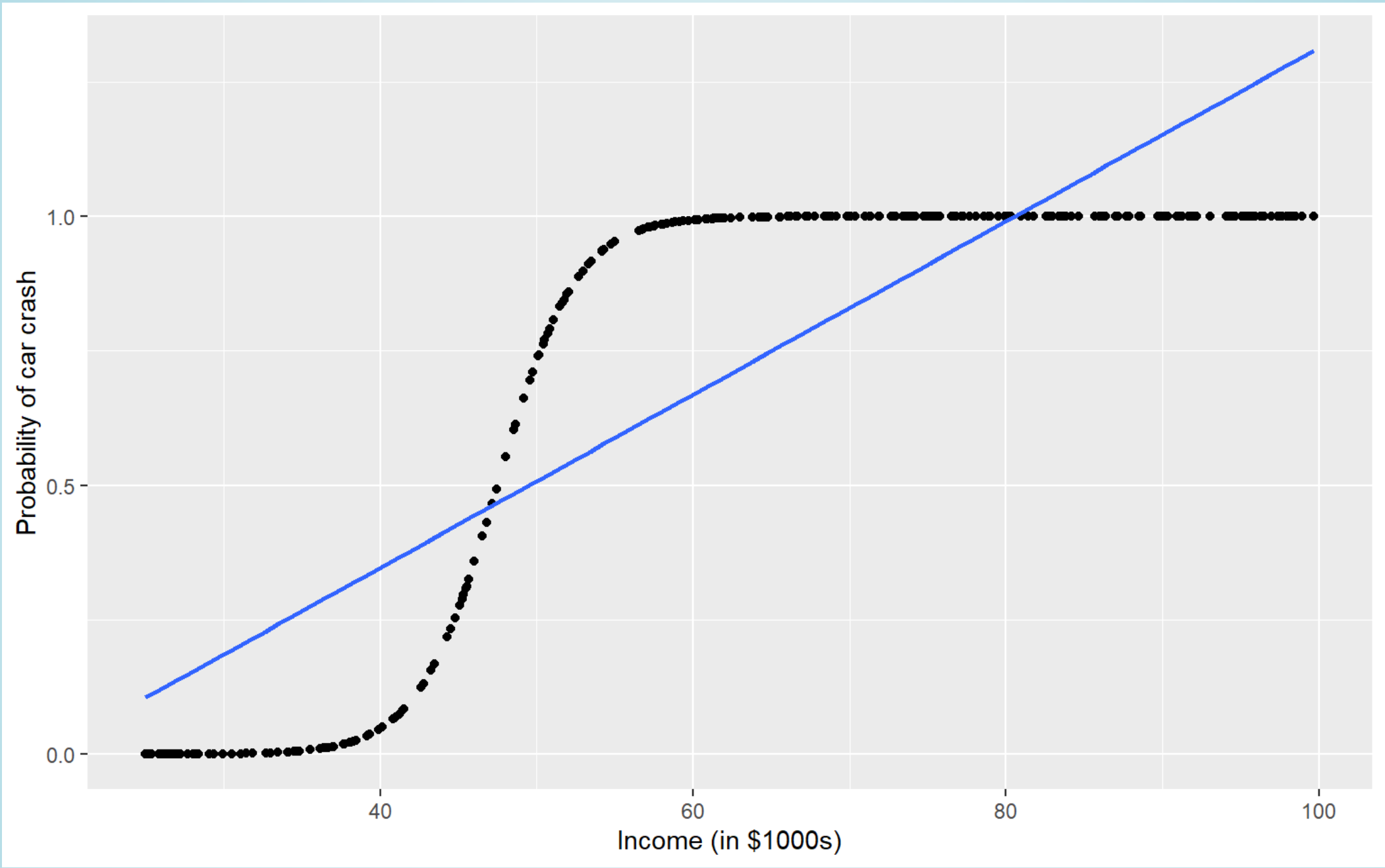
Let’s simulate data…



What does it mean to fit a line to this data?



We don’t get an error, but …


Probability of car crash
1.0
0.5
0.0
40
60
80
100
Income (in $1000s)

What is our expecation for people with an income of $45,000?

```r
crash_model <- lm(car_crash ~ income, data = sim_crash_data)

new_data <- data.frame(income = 45)

predict(crash_model, newdata = new_data)
```

```
        1
0.4151667
```

What is our expecation for people with an income of $90,000?

```r
crash_model <- lm(car_crash ~ income, data = sim_crash_data)

new_data <- data.frame(income = 90)

predict(crash_model, newdata = new_data)
```

```
       1
1.151587
```

Linear regression assumes

$$E[Y_i|X_i] = \beta_0 + \beta_1 X_{1,i} + \ldots + \beta_p X_{p,i} + \epsilon_i \text{ and } \epsilon_i \sim N(0, \sigma^2)$$

so

$$E[Y_i|X_{1,i}, \ldots, X_{p,i}] = \beta_0 + \beta_1 X_{1,i} + \ldots + \beta_p X_{p,i}$$

What is the expectation of a binary variable?

We can fit a linear model for a binary response, but:

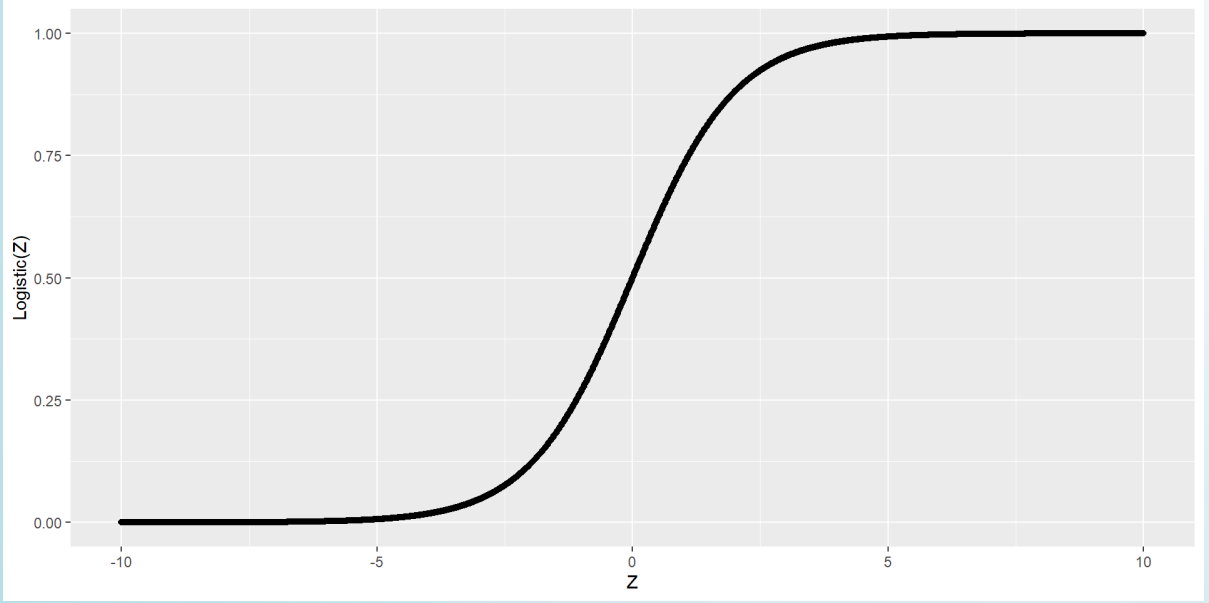
- the line is really fitting the $E[Y_i|X_i]$, the probability
- the range of $E[Y_i|X_i] = \beta_0 + \beta_1 X_{1,i} + \ldots + \beta_p X_{p,i}$ is $(-\infty, \infty)$
- we may get invalid predictions

We can use a logistic model instead!

Logistic function

The logistic function ranges from 0 to 1

$$\text{logistic}(Z) = \frac{e^Z}{1 + e^Z}$$



Logistic regression: form 1

Let $Y_i|X_{1,i},\ldots,X_{p,i} \sim Bernoulli(\pi_i)$, which means $E[Y_i|X_{1,i},\ldots,X_{p,i}] = \pi_i$

The logistic regression model is

$$E[Y_i|X_{1,i},\ldots,X_{p,i}] = \pi_i = \frac{e^{\beta_0+\beta_1 X_{1,i}+\ldots+\beta_p X_{p,i}}}{1+e^{\beta_0+\beta_1 X_{1,i}+\ldots+\beta_p X_{p,i}}}$$

Logistic Response Function Rewritten

$$\pi_i = \frac{e^{\beta_0+\beta_1 X_{1,i}+\ldots+\beta_p X_{p,i}}}{1+e^{\beta_0+\beta_1 X_{1,i}+\ldots+\beta_p X_{p,i}}}$$

Logistic regression: form 2

Through some algebra we can rewrite the model as

$$\log(\frac{\pi_i}{1-\pi_i}) = \beta_0 + \beta_1 X_{1,i} + \ldots + \beta_p X_{p,i}$$

- Form 1 is useful for understanding why we use the logistic function
- Form 2 is useful for interpretation

Logistic regression: form 2

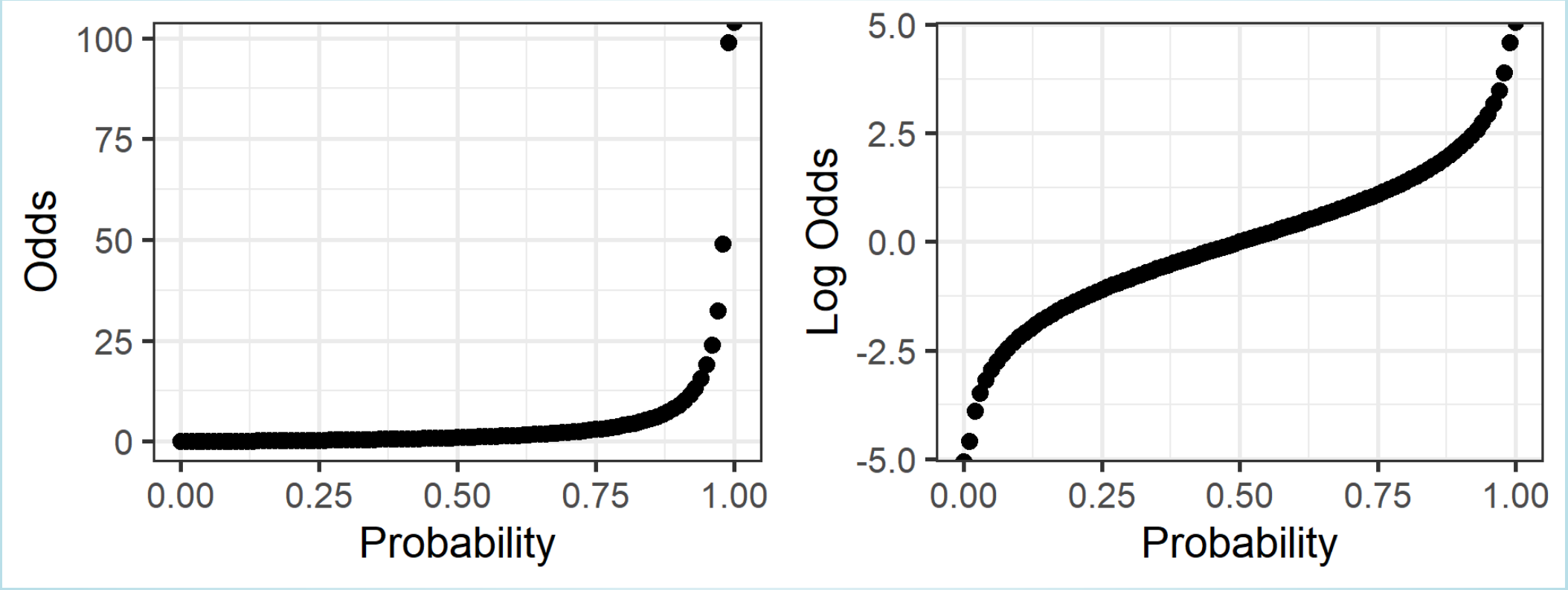
$$\log(\frac{\pi_i}{1-\pi_i}) = \beta_0 + \beta_1 X_{1,i} + \ldots + \beta_p X_{p,i}$$

- Left side is called the *log odds* or *logit*

Probability and Odds

What are odds?

Relationship between probability and odds



Parameter Interpretation: Solving for β_1

$$\log(\frac{\pi}{1-\pi}) = \beta_0 + \beta_1 X_1 + \ldots + \beta_{p-1} X_{p-1}$$

Now we have a model that connects $E[Y|X] = \pi$ to our data, but how do we interpret the parameters, β_j's?

Recall for linear regression: $E[Y|X] = \beta_0 + \beta_1 X$

Parameter Interpretation: Solving for β_1

$$\log\left(\frac{E[Y_i|X_i]}{1 - E[Y_i|X_i]}\right) = \beta_0 + \beta_1 X_i$$

Parameter Interpretation: Explaining β_1

Odds Ratio

$$e^{\beta_1} = \frac{\pi_a}{1-\pi_a} / \frac{\pi_b}{1-\pi_b}$$

We typically interpret the odds ratio e^{β_1} instead of log odds ratio β_1

If $e^{\beta_1} = 1$

If $e^{\beta_1} < 1$

If $e^{\beta_1} > 1$

Obtaining estimates

- For simple linear regression we can directly solve for the estimators
- For logistic regression we can’t directly calculate the estimators, we have to use optimization algorithms. Luckily those are implemented for us!

Inference With Logistic Regression

In linear regression we used a t-test to test for statistically significant features.

Beware in logistic regression there are different tests that can be used (we will talk about these)

- Likelihood Ratio Test
- Wald Test

Know which one your software is using!

Logisitic regression

$$\log\left(\frac{E[Y_i|X_{1i},\ldots,X_{pi}]}{1-E[Y_i|X_{1i},\ldots,X_{pi}]}\right) = \beta_0 + \beta_1 X_{1i} + \ldots + \beta_p X_{pi}$$

We often let $\pi_i = E[Y_i|X_{1i},\ldots,X_{pi}]$ so our model is written

$$\log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_0 + \beta_1 X_{1i} + \ldots + \beta_p X_{pi}$$

Interpreting coefficients

β_0 is the log odds for when all X_i's are zero.

β_1 is the log odds ratio associated with a one unit increase in X_1

- e^{β_1} is the odds ratio associated with a one unit increase in X_1
- e^{β_1} is the multiplicative change / factor change in odds associated with a one unit increase in X_i
- For $e^{\beta_1} > 1$, $100 * (e^{\beta_1} - 1)$% is the increase in odds associated with a one unit increase in X_1. For $e^{\beta_1} < 1$, we have $100 * (1 - e^{\beta_1})$% decrease in odds associated with a one unit increase in X_1

... with all other X_i's held constant.

Inference

In our class we will focus on two common methods for hypothesis tests and confidence intervals

- Wald
- Likelihood ratio (preferred for this class to avoid the instability of the Wald)

Coronary heart disease (cdh) data

A retrospective sample of males in a heart-disease high-risk region of the Western Cape, South Africa. These data are taken from a larger dataset, described in Rousseauw et al, 1983, South African Medical Journal.

- `sbp` systolic blood pressure
- `tobacco` cumulative tobacco (kg)
- `ldl` low density lipoprotein cholesterol
- `adiposity` body fat
- `famhist` family history of heart disease
- `typea` type-A behavior
- `obesity`
- `alcohol` current alcohol consumption
- `age` age at onset

```r
heart_data <- read_csv(
  "https://hastie.su.domains/ElemStatLearn/datasets/SAheart.data",
  show_col_types = FALSE
)

glimpse(heart_data)
```

```
Rows: 462
Columns: 11
$ row.names <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 1…
$ sbp       <dbl> 160, 144, 118, 170, 134, 132, 142, 114, 114, 132, 206, 134, …
$ tobacco   <dbl> 12.00, 0.01, 0.08, 7.50, 13.60, 6.20, 4.05, 4.08, 0.00, 0.00…
$ ldl       <dbl> 5.73, 4.41, 3.48, 6.41, 3.50, 6.47, 3.38, 4.59, 3.83, 5.80, …
$ adiposity <dbl> 23.11, 28.61, 32.28, 38.03, 27.78, 36.21, 16.20, 14.60, 19.4…
$ famhist   <chr> "Present", "Absent", "Present", "Present", "Present", "Prese…
$ typea     <dbl> 49, 55, 52, 51, 60, 62, 59, 62, 49, 69, 72, 65, 59, 49, 54, …
$ obesity   <dbl> 25.30, 28.87, 29.14, 31.99, 25.99, 30.77, 20.81, 23.11, 24.8…
$ alcohol   <dbl> 97.20, 2.06, 3.81, 24.26, 57.34, 14.14, 2.62, 6.72, 2.49, 0.…
$ age       <dbl> 52, 63, 46, 58, 49, 45, 38, 58, 29, 53, 60, 40, 17, 15, 53, …
$ chd       <dbl> 1, 1, 0, 1, 1, 0, 0, 1, 0, 1, 1, 1, 0, 0, 0, 0, 0, 1, 1, 1, …
```

Fitting the model

$$\log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_0 + \beta_1 X_{\text{Blood pressure},i} + \beta_2 X_{\text{Family history},i} + \beta_1 X_{\text{Age},i}$$

```r
heart_model <- glm(
  chd ~ sbp + famhist + age,
  data = heart_data,
  family = binomial # This tells it to do logisitic regression
)
```

```r
summary(heart_model)
```

```
Call:
glm(formula = chd ~ sbp + famhist + age, family = binomial, data = heart_data)

Coefficients:
                Estimate Std. Error z value Pr(>|z|)
(Intercept)    -4.500388   0.767490  -5.864 4.52e-09 ***
sbp             0.006404   0.005389   1.188    0.235
famhistPresent  0.943106   0.216915   4.348 1.37e-05 ***
age             0.056222   0.009239   6.086 1.16e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)
```

Testing example 1

Let's test if systolic blood pressure should be in the model

$H_0 : \beta_1 = 0$

$H_A : \beta_1 \neq 0$

We can do this with a Wald or Likelihood ratio test

```
heart_full_model <- glm(
  chd ~ sbp + famhist + age,
  data = heart_data,
  family = binomial # This tells it to do logisitic regression
)
```

Performing Wald Test

```
1  summary(heart_full_model)
```

```
Call:
glm(formula = chd ~ sbp + famhist + age, family = binomial, data = heart_data)

Coefficients:
                Estimate Std. Error z value Pr(>|z|)
(Intercept)    -4.500388   0.767490  -5.864 4.52e-09 ***
sbp             0.006404   0.005389   1.188    0.235
famhistPresent  0.943106   0.216915   4.348 1.37e-05 ***
age             0.056222   0.009239   6.086 1.16e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)
```

The `summary()` function displays Wald Statistics.

Performing Likelihood Ratio Tests

```r
heart_reduced_model <- glm(
  chd ~ famhist + age,
  data = heart_data,
  family = binomial # This tells it to do logisitic regression
)

anova(heart_reduced_model, heart_full_model, test = "LRT")
```

```
Analysis of Deviance Table

Model 1: chd ~ famhist + age
Model 2: chd ~ sbp + famhist + age
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1       459     506.66
2       458     505.24  1    1.414   0.2344
```

- The p-value for Wald test: 0.235
- The p-value for the Likelihood Ratio Test: 0.2344
- The two p-values won't necessarily equal the same. In practice you would choose one test to perform, you wouldn't fit both.

A probability of 0.02344 is not unlikely, so fail to reject H_0.

We found no evidence (Likelihood ratio test p-value = 0.2344) that the model with systolic blood pressure significantly improves the fit compared to the model with just family history and age.

Testing example 2

Let's test if systolic blood pressure and family history should be in the model

$H_0 : \beta_1 = \beta_2 = 0$

H_A : At least one of β_1 or β_2 are zero

We can do this with a Wald or Likelihood ratio test

Wald test

```r
library(aod)

wald.test(
  Sigma = vcov(heart_full_model),
  b = coef(heart_full_model),
  Terms = c(1, 2)
)
```

```
Wald test:
----------

Chi-squared test:
X2 = 75.0, df = 2, P(> X2) = 0.0
```

So we can test subsets of predictors with a Wald test, but it is different...

Likelihood ratio test

```r
heart_reduced_model2 <- glm(
  chd ~ age,
  data = heart_data,
  family = binomial
)

anova(heart_reduced_model2, heart_full_model, test = "LRT")
```

```
Analysis of Deviance Table

Model 1: chd ~ age
Model 2: chd ~ sbp + famhist + age
  Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
1       460     525.56
2       458     505.24  2   20.318 3.872e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Let’s use the likelihood ratio test.

We found very strong evidence (likelihood ratio test p-value 3.9e-05) that including systolic blood pressure and family history improve the model fit, over a model with age only.

Confidence intervals

Via Likelihood Ratio: We can used the `confint()` function like we did with linear regression

```
1  confint(heart_full_model)
```

```
                    2.5 %      97.5 %
(Intercept)    -6.038659314 -3.02234201
sbp            -0.004159741  0.01703609
famhistPresent  0.520038741  1.37140982
age             0.038573599  0.07486714
```

Via Wald: We can used the `confint.default()` function

```
1  confint.default(heart_full_model)
```

```
                    2.5 %      97.5 %
(Intercept)    -6.004640777 -2.99613599
sbp            -0.004158755  0.01696642
famhistPresent  0.517961224  1.36825057
age             0.038114783  0.07432913
```